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Ronan Querrec (Lab-STICC)
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Subject: Application for PhD Position – Modeling Interaction Scenarios for Digital Twins in Mixed Reality (PC5)

Dear Ronan Querrec, Valérie Gouranton, Vincent Havard, and Eloïdie Bouzekri,

I am applying for the PC5 PhD position on "Modeling Interaction Scenarios for Digital Twins in Mixed Reality" within the EDT program. Combining human-in-the-loop interaction models with high-fidelity digital replicas requires both robust software architecture and intuitive real-time visualizations—challenges I am eager to tackle.

Through my project on the "Live Control Room," I architected a high-performance 3D web dashboard (Next.js/Three.js) capable of rendering complex simulated environments and interacting dynamically with underlying AI models. This required me to design structured interaction scenarios where user inputs cleanly interfaced with the physical twin's data streams. I am confident that my experience in bridging graphical representation and data synchronization will allow me to significantly advance the state of the art in Mixed Reality scenarios, ensuring that business logic and interactive behaviors remain coupled in real-time. By applying model-driven engineering principles, we can generalize interaction scenarios across various MR devices.

I am particularly excited about the opportunity to validate this research on real-world industrial platforms and contribute to the Artemis framework. Please find attached my resume and academic transcripts. I look forward to discussing how my technical competencies align with the laboratories and the broader EDT program goals.

Sincerely,

Ismail Aissaoui